Association between dietary meat consumption and incident type 2 diabetes: the EPIC-InterAct study.

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AIMS/HYPOTHESIS: A diet rich in meat has been reported to contribute to the risk of type 2 diabetes. The present study aims to investigate the association between meat consumption and incident type 2 diabetes in the EPIC-InterAct study, a large prospective case-cohort study nested within the European Prospective Investigation into Cancer and Nutrition (EPIC) study. METHODS: During 11.7 years of follow-up, 12,403 incident cases of type 2 diabetes were identified among 340,234 adults from eight European countries. A centre-stratified random subsample of 16,835 individuals was selected in order to perform a case-cohort design. Prentice-weighted Cox regression analyses were used to estimate HR and 95% CI for incident diabetes according to meat consumption. RESULTS: Overall, multivariate analyses showed significant positive associations with incident type 2 diabetes for increasing consumption of total meat (50 g increments: HR 1.08; 95% CI 1.05, 1.12), red meat (HR 1.08; 95% CI 1.03, 1.13) and processed meat (HR 1.12; 95% CI 1.05, 1.19), and a borderline positive association with meat iron intake. Effect modifications by sex and class of BMI were observed. In men, the results of the overall analyses were confirmed. In women, the association with total and red meat persisted, although attenuated, while an association with poultry consumption also emerged (HR 1.20; 95% CI 1.07, 1.34). These associations were not evident among obese participants. CONCLUSIONS/INTERPRETATION: This prospective study confirms a positive association between high consumption of total and red meat and incident type 2 diabetes in a large cohort of European adults.